SOFTWARE ENGINEERING PROJECT

KUru: AI-Powered University Program Advisor for KU Program Selection using RAG and Semantic Recommendation (Proposal)

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Chapter 1

Introduction

1.1 Background

Thailand's university admission system, TCAS (Thai University Central Admission System), requires students to select and apply to specific faculties at specific universities through up to four rounds, each with distinct requirements, quotas, and criteria. Kasetsart University (KU), one of Thailand's leading public universities, offers programs spanning agriculture, engineering, science, humanities, business, and more across its campuses.

Despite the availability of official information about each program, most prospective students make faculty selection decisions with limited structured guidance. Program Learning Outcomes (PLOs), which formally document the competencies a graduate will have developed, are defined for every accredited program in Thailand via the TQF (Thai Qualifications Framework) document system, known as มคอ.2. These documents are the authoritative source of what a program actually teaches and produces, yet they are inaccessible to the students they most directly affect: written in formal Thai, formatted inconsistently across faculties, and distributed as PDFs on faculty websites without any structured interface.

Prospective students who are interested in KU programs — whether finishing high school, taking a gap year, or reconsidering their options — face this problem acutely, as they must make high-stakes decisions with limited structured guidance.

1.2 Problem Statement

High school students selecting KU programs for TCAS lack personalized, grounded guidance that connects their interests to specific program learning outcomes, curriculum content, and admission requirements. Existing tools are either generic (general-purpose LLMs with no KU-specific data) or static (official faculty web pages with no personalization or synthesis).

The consequence is suboptimal faculty selection, contributing to higher rates of program dissatisfaction, transfer requests, and dropout.

1.3 Solution Overview

KUru is a bilingual (Thai and English) web application that serves as an intelligent advisor for students who are exploring programs at Kasetsart University. The system is grounded in official KU curriculum documents provided by faculty, making its guidance specific, accurate, and authoritative rather than approximate.

The target system conducts adaptive interest discovery, produces ranked program recommendations with PLO visualizations, and presents structured TCAS admission information for matched programs. The current proof of concept is narrower: it demonstrates data ingestion, the grounded RAG chatbot, and the Program Explorer.

The system focuses exclusively on students at the pre-admission stage — before any enrollment decision is made.

The long-term KUru architecture combines a RAG pipeline over มคอ.2 curriculum documents with structured recommendation, curriculum, admission, and future graph-based reasoning components. For the evaluated MVP, the system is intentionally narrower: it proves grounded curriculum and TCAS question answering, RIASEC-based interest discovery, semantic program search, simplified program recommendation, static PLO visualisation, and recommendation explanation. More advanced graph-based, behavioural, comparison, career, portfolio, and timeline features are retained as Phase 2 target-system capabilities.

1.4 Revision Based on Proposal Feedback

The proposal committee's main concern was recommendation evaluation: whether a student truly "liked" or benefited from a program can only be known after years of study, so long-term satisfaction is not a feasible MVP ground truth. This revised SRS addresses that concern by removing career-path validation from the evaluated MVP and replacing it with short-term, inspectable proxy evaluation. The MVP recommendation is evaluated by whether ranked programs are relevant to a student's stated interests and retrieved curriculum evidence, using a curated profile test set, advisor/senior-student relevance labels, MRR/NDCG@5, explanation faithfulness, and user task feedback. Career Explorer, O*NET matching, Neo4j graph reasoning, and behavioural re-ranking are retained only as Phase 2 target-system features.

Other proposal feedback is addressed through three additional changes. First, the MVP scope is reduced from the original full-product vision to the evaluated features that can be implemented and tested reliably. Second, the recommendation architecture is split into an MVP implementation using Pipeline B2 semantic curriculum matching and a Phase 2 target architecture. Third, RAG evaluation is clarified by separating the May 2026 POC custom LLM-as-judge regression tests from the final MVP RAGAS targets, while the current POC status is documented explicitly as completed technical evidence rather than final MVP success.

1.5 MVP Scope Boundary

The original KUru product vision includes a broad AI pathway navigator. The evaluated MVP narrows that vision into a smaller system that can be implemented and measured reliably within the project period. Table 1.1 defines the source of truth for MVP scope used throughout this SRS.

Table 1.1: MVP Scope Boundary

Status	**Use Cases / Features**	**MVP Interpretation**
In scope	UC-01 Interest Discovery; UC-02 Program Recommendation; UC-04 KUru Advisor; UC-08 Recommendation Explanation; UC-09 Semantic Program Search	Core evaluated MVP capabilities. Recommendation is simplified to curriculum-side semantic matching using Pipeline B2 rather than the full hybrid target architecture.
Partial scope	UC-03 PLO Chart; UC-05 TCAS Admission Guide	Implemented as static PLO/profile visualisation and TCAS information on program detail pages and chatbot answers. A full interactive PLO overlay and standalone TCAS guide are Phase 2.
Phase 2 / target system	UC-06 Career Explorer; UC-07 Behavioural Blending; UC-10 Program Comparison; UC-11 Curriculum Timeline Visualiser; UC-12 Portfolio Readiness Checker; Neo4j Pipeline B1	Preserved in the product vision but excluded from MVP pass/fail evaluation. Prototype UI may exist, but these are not grading targets for the evaluated MVP.

1.6 Current POC Baseline

As of May 2026, the current proof of concept demonstrates the technical baseline for the MVP rather than the complete target system. The POC includes a cleaned Bangkhen-focused corpus of 57 programs and 13,910 indexed chunks, the data ingestion pipeline, a structured v8 RAG pipeline selected for TCAS and fee grounding, source-cited chatbot responses, a Program Explorer over indexed curriculum content, frontend/backend integration, and MLflow-backed evaluation. The selected production regression run, `v8_structured_tcas_fees`, records 72.7% good answers and an average score of 2.055/3.0 on the fee/TCAS regression suite. The headline retrieval benchmark, `latest_submission_headline_v7_rerank_74pct`, records 74% good answers and an average score of 2.26/3.0. The current retrieval configuration uses Supabase pgvector with local `intfloat/multilingual-e5-base` embeddings, `top_k = 5`, `min_similarity = 0.35`, and `fetch_multiplier = 3`. Known POC limitations include dense tables, OCR-noisy chunks, incomplete campus coverage, a DVM record (`bangkhen_vet_9ea52f52`) with 0 chunks after an interrupted large-PDF retry, non-clickable source chips, RAGAS not yet serving as the primary metric, and final recommender metrics not yet implemented. Interest discovery, recommendation, recommendation explanation, and PLO/profile visualisation remain final MVP work to be completed and evaluated against the MVP evaluation plan.

1.6.1 Features

The list below preserves the broader target-system feature set while labelling each item by its evaluated MVP status.

1. **Interest Discovery (MVP):** A twelve-step adaptive interest elicitation interface grounded in Holland's RIASEC vocational model [1]: (1) six sequential Likert rating screens, one per RIASEC dimension (R/I/A/S/E/C), each presenting 4 Thai-language statements rated on a 5-point scale

(1 = ไม่เห็นด้วยอย่างมาก, 5 = เห็นด้วยอย่างมาก) — raw scores per dimension are summed (max 20); (2) one global confidence check screen applying a scalar (1.0 / 0.75 / 0.5) to all six dimension scores; (3) 4–6 adaptive pairwise forced-choice questions targeting only dimension pairs where $|\text{score}_A - \text{score}_B| < 3$ after confidence scaling; (4–6) 3 scenario questions with A–F role options mapped to RIASEC dimensions, adjusting scores proportionally; and (7) a profile summary page with an optional dealbreaker filter zeroing out explicitly rejected dimensions before L2-normalisation. Total elicitation time is 2–4 minutes. Behavioural refinement is deferred to Phase 2.

2. **Program Recommendation Engine (MVP simplified):** Ranked list of KU programs matched to the student's RIASEC interest profile. The evaluated MVP uses curriculum-side semantic matching over indexed program and course content (Pipeline B2 only). The target-system architecture combines a career-side signal (Pipeline A), Neo4j PLO matching (Pipeline B1), and course-content semantic search (Pipeline B2), with final score $= 0.35 \times$ A-score $+ 0.65 \times$ B-score. The full hybrid architecture is deferred to Phase 2.

3. **PLO Spider Chart Visualizer (Partial MVP):** Static chart showing the skill profile a student will develop in any given program. Interactive overlay with the student's interest profile is a Phase 2 enhancement.

4. **KUru Advisor (MVP):** RAG-powered Q&A interface over indexed curriculum, TCAS, and fee-related documents supporting natural language queries in Thai and English, with source citations and explicit missing-data fallback behaviour.

5. **TCAS Admission Guide (Partial MVP):** Program detail pages and chatbot answers expose available TCAS infor-

mation including GPAX requirements, exam score criteria (TGAT/TPAT/A-Level), portfolio requirements, fees, and deadlines. A standalone TCAS guide is Phase 2.

6. **Saved Profile Dashboard (Phase 2):** Optional login to persist interest profile, bookmarked programs, and TCAS deadline tracking across sessions.

7. **PLO Explorer with Semantic Search (MVP):** A browsable and searchable directory of indexed KU programs. Students can search using natural language queries in Thai or English (e.g., "หลักสูตรที่เรียน AI เน้นโปรเจกต์ รับ TGAT ต่ำ"). The MVP matches against program content via Supabase/pgvector and available metadata. Multi-source Neo4j search, pin-tray comparison, and batch portfolio checking are Phase 2.

8. **Portfolio Readiness Checker (Phase 2):** Students upload a portfolio PDF; Gemini extracts a structured portfolio profile (activities, certificates, academic records, personal statement). The system compares this against pre-extracted faculty criteria schemas for selected programs and performs a four-part gap analysis. Prototype UI may exist, but this feature is excluded from MVP pass/fail evaluation.

9. **Curriculum Timeline Visualiser (Phase 2):** For any KU program, displays a visual map of what the student will study across 4 years. Course composition, teaching method breakdown (lecture/lab/project ratios), and time commitment per year are extracted from มคอ.2 course structure data during ingestion.

10. **Program Comparison (Phase 2):** Students can pin up to 4 programs from the explorer or recommendation screen and compare them side-by-side, showing: overlaid PLO radar charts, career path overlap, curriculum character profile across 5 dimensions, TCAS accessibility per round, and

personalised fit scores relative to the student's RIASEC profile.

1.7 Target Users

KUru serves prospective students at the pre-admission stage:

- **Prospective KU applicants (pre-admission):** Anyone considering KU programs before making an enrollment decision. This includes high school students in Mathayom 4–6 preparing for TCAS, gap-year students reconsidering their options, and anyone exploring what KU programs offer through curriculum, PLO, and admission evidence. No prior KU enrollment is assumed. Guidance spans from initial interest discovery through to understanding specific TCAS admission requirements. Career-path exploration is retained as a Phase 2 target-system capability.

Users may access the MVP without an account. Login-based profile persistence, saved programs, and deadline notifications are Phase 2 enhancements unless time remains after MVP evaluation.

1.8 Terminology

Table 1.2: Terminology

Term	**Definition**
PLO	Program Learning Outcome — a formally defined competency that graduates of a specific program are expected to have developed, as documented in มคอ.2.
CLO	Course Learning Outcome — a learning outcome defined at the individual course level, documented in มคอ.3. CLOs contribute to PLOs.
มคอ.2	The Thai Qualifications Framework program specification document for each accredited university program. Contains PLOs, curriculum structure, and admission requirements.
มคอ.3	The course-level specification document detailing CLOs, teaching methods, and assessment for each course.
TCAS	Thai University Central Admission System — the national university admission platform through which students apply to Thai universities in up to four rounds per year.
TGAT/TPAT/A-Level	Standardized national examinations used as admission criteria in TCAS Rounds 2–4.
RAG	Retrieval-Augmented Generation — an AI architecture that retrieves relevant passages from a document corpus and provides them as context to a language model before generating a response, grounding output in retrieved evidence.
Knowledge Graph	A graph database structure in which nodes represent entities (faculties, PLOs, skills, careers) and edges represent relationships between them (a faculty produces certain PLOs, a PLO develops certain skills, a career requires certain skills).
O*NET	The Occupational Information Network — a US Department of Labor database cataloguing approximately 900 occupations with structured skill requirements. Used in the Phase 2 career-to-skill data layer.
RIASEC	Holland's six vocational personality types: Realistic, Investigative, Artistic, Social, Enterprising, Conventional. Used as the theoretical framework for interest elicitation and as the native classification system of the O*NET occupational database.
Holland Code	A two-letter code formed from a person's two most dominant RIASEC types (e.g., IR for Investigative-Realistic), used as a human-readable summary of an individual's vocational interest profile.

Chapter 2

Literature Review and Related Work

2.1 Competitor Analysis

Five categories of existing solutions are relevant to this project:

2.1.1 KU Official Information Channels

KU's official website and faculty pages provide static information about programs, curricula, and admission requirements. PLO information is available only within มคอ.2 PDF documents, which are distributed without structured interfaces. TCAS information is scattered across the main KU admissions page, individual faculty pages, and the national mytcas.com platform. No personalization, synthesis, or guidance is offered. The fundamental limitation is that information exists but is not actionable for students making decisions.

2.1.2 General-Purpose LLMs

General-purpose language models such as ChatGPT and Gemini can respond to natural language queries about KU programs, but do so from training data rather than authoritative sources. This results in approximations and hallucinations when asked about specific PLOs, current TCAS requirements, or quota numbers. These models have no persistent memory of the student across sessions and provide no structured outputs such as visualizations or deadline trackers.

2.1.3 University-Specific AI Advisors at Other Thai Institutions

Knowva (KMITL) and ChulaGENIE (Chulalongkorn University) represent the closest competitors to KUru. Both are AI-assisted advising systems built for specific Thai universities. However, neither is available at KU, neither explicitly links PLOs to career skill requirements, and neither provides TCAS-specific admission guidance. ChulaGENIE focuses on course recommendation for enrolled students rather than pre-admission guidance.

2.1.4 TCAS Information Platforms

Dedicated TCAS information platforms, most prominently mytcas.com (the official national TCAS portal operated by the Ministry of Higher Education), aggregate admission data across all Thai universities into a single structured interface. For each program, mytcas.com presents per-round quotas, eligibility criteria (curriculum type, GPAX thresholds), score calculation methods (portfolio weighting, TGAT/TPAT/A-Level proportions), and links to faculty admission pages. The data is updated each TCAS cycle and is the authoritative source consulted by both students and universities.

Despite its comprehensiveness, mytcas.com is a passive information browser, not a guidance system. It presents raw admission data without connecting it to a student's interests, skills, or career goals. Students must independently navigate from a program they have already identified to its admission requirements; the platform provides no discovery mechanism, no program matching, and no PLO or career-path context. KUru is designed to complement mytcas.com by providing the personalized discovery and program-matching layer that sits upstream of the admission details mytcas.com supplies.

2.1.5 Comparison Summary

Table 2.1 summarises how KUru addresses the gaps identified across the five categories of existing solutions.

Table 2.1: Competitor Comparison Summary

Feature	KU Website	mytcas .com	ChatGPT/ Gemini	Knowva/ Chula-GENIE	KUru (ours)
KU-specific curriculum data	Partial (PDF)	✗	✗	✗	Full (มคอ.2 RAG)
PLO visualization	✗	✗	✗	✗	Spider chart
Career-to-PLO mapping	✗	✗	∼	∼	Phase 2 graph
TCAS admission guide	Scattered	Full — all universities	∼	✗	Structured, KU only
Personalization	✗	✗	Session-only	Basic	RIASEC profile; progressive Phase 2
Thai language support	✓	✓	∼	∼	✓
KU student skill tracking	✗	✗	✗	✗	Yes (Phase 2)

2.2 Literature Review

2.2.1 Retrieval-Augmented Generation (RAG)

Lewis et al. [2] introduced RAG as an architecture combining dense retrieval with sequence-to-sequence generation for open-domain question answering. The key insight is that grounding generation in retrieved evidence substantially reduces hallucination and allows the model to answer questions requiring specific factual knowledge not reliably encoded in model weights. This is directly applicable to KUru's requirement to answer questions about specific PLO content, admission criteria, and course structures from มคอ.2 documents.

For evaluation, the RAGAS framework [3] provides automated metrics for RAG pipeline quality including faithfulness, answer relevancy, and context recall, enabling objective assessment of whether the system's answers are grounded in retrieved curriculum content.

2.2.2 Knowledge Graphs for Educational Recommendation

Knowledge graph-based recommendation systems have been demonstrated in educational contexts to enable multi-hop reasoning that purely embedding-based approaches cannot support. Embedding-based retrieval can identify semantically similar items but cannot follow explicit typed relationships such as ‘this PLO develops this skill’ or ‘this career requires this skill level.’ The Neo4j graph in KUru encodes these typed relationships explicitly, enabling recommendation paths that are both explainable and traceable to specific curriculum documents [4, 5].

Using knowledge graphs as context sources for LLM-generated explanations has been specifically investigated by Abu-Rasheed et al. [6], who propose a conversational explainability approach where knowledge graph data is injected into LLM prompts to reduce hallucination risk while maintaining natural dialogue. This is the target-system architectural pattern KUru adopts for Phase 2: Neo4j provides the structured PLO-to-skill data, and Gemini generates the natural-language explanation from that data rather than from general knowledge. The MVP uses retrieved curriculum evidence from Supabase/pgvector as the explanation source.

2.2.3 Adaptive Quiz and Interest Profiling

KUru’s interest discovery interface is grounded in Holland’s RIASEC model (Realistic, Investigative, Artistic, Social, Enterprising, Conventional), the dominant framework in vocational psychology for connecting individual interests to occupational environments [1]. This theoretical foundation is directly compatible with KUru’s O*NET career data layer [7], which uses RIASEC codes to classify all 900+ occupations. Rather than presenting abstract RIASEC labels, KUru presents six sequential Likert screens using relatable Thai-language statements grounded in Holland’s six dimensions, reducing cognitive load for a high school student audience while producing a more reliable initial vector than a tile-selection approach. The elicitation design is defined in Section 4.3.2.

The interface design is shaped by three evidence-based principles from the preference elicitation (PE) and recommender systems literature.

First, the system uses explicit attribute-based elicitation rather than item-based elicitation. Research on large-scale onboarding systems demonstrates that item-based approaches — asking users to evaluate individual items directly — are unsuitable when the item corpus is large, because no single item efficiently maps a new user’s preference space [8]. KUru instead presents topic clusters (attributes) that map directly to

RIASEC dimensions and downstream skill vectors, enabling broad preference coverage in minimal interactions.

Second, the questionnaire is deliberately minimal. Sepliarskaia et al. [9] demonstrated at ACM RecSys that an optimized Static Preference Questionnaire (SPQ) can reduce necessary questionnaire length by up to a factor of three compared to standard elicitation methods, while maintaining or improving recommendation quality. While the SPQ principle advocates for minimalism, KUru's five-step design remains substantially shorter than validated instruments while incorporating adaptive question selection — targeting only ambiguous RIASEC dimensions rather than administering all questions uniformly — which preserves the efficiency principle while improving signal quality. KUru's twelve-step adaptive design — (1) six Likert screens (4 statements per dimension, 5-point scale); (2) one global confidence check; (3) 4–6 adaptive pairwise questions targeting only ambiguous dimension pairs (delta $<$ 3 points); (4) three scenario questions confirming the dominant signal; (5) an optional dealbreaker filter on the profile summary page — is substantially shorter than a full validated interest inventory while producing a more reliable initial vector than a single tile-selection approach. The pairwise comparison step is adaptive: students with a clearly differentiated profile encounter fewer questions, keeping total completion time within 2–4 minutes.

Third, the system addresses the new user cold-start problem [10] through this explicit elicitation phase. Progressive implicit profiling from behavioral signals such as program page visits, chatbot queries, and saved items is retained as Phase 2 [11, 12]. The structured grid interface is supported by de Vries et al. [13], who found in a controlled user study that high-guidance elicitation — presenting structured options rather than open-ended prompts — produces significantly higher recommendation match scores than low-guidance alternatives.

It should be noted that interest-based elicitation has inherent limitations as a predictor of long-term academic fit. Meta-analytic research on full validated interest inventories reports approximately 50% accuracy in predicting eventual career choice even with substantially longer instruments [1]. KUru's abbreviated elicitation is accordingly framed as a starting point for exploration rather than a definitive match. The evaluated

MVP therefore measures short-term relevance and explanation faithfulness, while later confidence improvements from implicit profiling are deferred to Phase 2.

This work applies established cold-start PE methods and the RIASEC vocational framework to academic pathway advising for Thai university applicants — a novel application context in which the user population (Thai high school students navigating TCAS), the item space (university programs defined by PLOs), and the decision stakes (a multi-year academic commitment) differ substantially from the content recommendation domains in which these methods were originally developed.

2.2.4 Hybrid Recommendation Architecture

KUru's target-system recommendation pipeline is a hybrid system combining two independent signals: a knowledge-based signal derived from matching the student's RIASEC vector against O*NET occupation profiles (Pipeline A), and a content-based signal derived from matching against KU program PLO profiles and course content via Neo4j and pgvector semantic search (Pipeline B). Hybrid recommender systems that combine independent evidence sources consistently outperform single-signal approaches by reducing the error compounding that occurs when one signal feeds sequentially into the next [10]. In the previous single-chain design, an approximation introduced at the O*NET-to-SkillCluster crosswalk step would propagate through to the final ranking; the parallel architecture eliminates this dependency. The MVP uses the curriculum-side semantic component (Pipeline B2) first, then evaluates the full hybrid architecture in Phase 2.

Chapter 3

Requirement Analysis

3.1 Stakeholder Analysis

- **High school students (prospective KU applicants):** Primary users. Need personalized program recommendations, PLO-based skill previews, curriculum evidence, and structured TCAS guidance. Low technical proficiency expected; interface must be accessible and engaging. Career-path information is treated as a Phase 2 extension.
- **KU Faculty / Academic Advisors:** Data providers. Provide มคอ.2 curriculum documents and TCAS admission data. Indirect beneficiaries through reduced advising load and better-prepared incoming students.
- **KU Admissions Office:** Indirect stakeholders. Benefit from better-informed applicants and reduced mismatched enrollments.
- **Development Team:** Two SKE students responsible for design, development, and evaluation within one semester.

3.2 User Stories and Use Cases

3.2.1 High School Student User Stories

- As a high school student, I want to discover which KU programs match my interests so that I can make an informed TCAS application.

- As a high school student, I want to see what skills I will graduate with from a specific program so that I can assess whether it aligns with my career goals.
- As a high school student, I want to understand the TCAS admission requirements for a program I am interested in so that I know what I need to prepare.
- As a high school student, I want to ask questions about a program's curriculum in Thai so that I can get specific answers without reading long PDF documents.
- As a high school student, I want to bookmark programs I am considering so that I can return to them later. Multi-program comparison is a Phase 2 target-system workflow.

3.2.2 Additional User Stories

- As a returning visitor, I want my saved RIASEC interest profile and bookmarked programs to be available when I come back so that I do not have to start over. Persistent login-based storage is a Phase 2 enhancement unless time remains after MVP evaluation.
- As a parent, I want to browse KU programs and understand their curriculum, PLOs, and admission requirements so that I can support my child's decision.

3.2.3 Key Use Cases

Table 3.1 classifies each use case by MVP evaluation status. Use cases marked Phase 2 remain part of the target product vision but are excluded from MVP pass/fail evaluation.

UC-01: Discover Interests
Actor: Guest / High School Student
Precondition: None — accessible without login.
Main Flow:

Table 3.1: Use Case MVP Status

Use Case	**Name**	**MVP Status**
UC-01	Discover Interests	In scope. Evaluated through completion rate and usability.
UC-02	View Program Recommendations	In scope, simplified to Pipeline B2 semantic curriculum matching. Pipeline A and Neo4j Pipeline B1 are Phase 2.
UC-03	Explore PLO Spider Chart	Partial. Static PLO/profile chart only in MVP. Student overlay and graph-derived skill mapping are Phase 2.
UC-04	Query KUru Advisor	In scope. Evaluated with RAGAS plus custom TCAS/fee regression checks.
UC-05	View TCAS Admission Guide	Partial. TCAS data appears in program detail pages and chatbot answers. Standalone guide and comparison are Phase 2.
UC-06	Explore Career Paths	Phase 2.
UC-07	Refine Profile Through Implicit Signals	Phase 2.
UC-08	Explain Why a Program Was Recommended	In scope. Explanation must cite the retrieved curriculum evidence used in the simplified recommendation.
UC-09	Search Programs Semantically	In scope. Natural-language semantic search over indexed program content and metadata.
UC-10	Pin Programs and Compare	Phase 2.
UC-11	View Curriculum Timeline	Phase 2.
UC-12	Portfolio Readiness Check	Phase 2.

1. Student opens KUru and navigates to ค้นหาความสนใจ. System presents six sequential screens, one per RIASEC dimension (R, I, A, S, E, C), each showing 4 Thai-language statements rated on a 5-point Likert scale (1 = ไม่เห็นด้วยอย่างมาก, 5 = เห็นด้วยอย่างมาก). A progress bar shows current step out of 12 total.

2. System computes a raw score per dimension (sum of 4 responses, max 20). Before proceeding to the pairwise step, system presents one confidence check screen: "คุณรู้สึกมั่นใจแค่ไหนกับคำตอบที่เพิ่งเลือก?" with three options (มั่นใจมาก = scalar 1.0, ค่อนข้างมั่นใจ = 0.75, ไม่แน่ใจเลย = 0.5). The scalar is applied globally to all six Likert-derived dimension scores.

3. System identifies ambiguous dimension pairs — any two dimensions where $|\text{score}_A - \text{score}_B| < 3$ after confidence scaling. Only pairs meeting this threshold are shown as pairwise forced-choice questions (maximum 4–6 pairs). Students with a clearly differentiated profile receive fewer questions. Each pair presents two options with "ชอบเท่ากัน" as a middle choice.

4. System presents 3 scenario questions (สถานการณ์ที่ 1–3), each offering A–F role options mapped to RIASEC dimensions. Student selects one preferred role per scenario. Scenario responses adjust dimension scores proportionally.

5. System displays the profile summary page showing the top 2 dominant dimensions, a bar chart of all 6 dimension scores (out of 20), and a Holland Code label. Below the summary, system presents an optional dealbreaker filter: "มีสาขาไหนที่คุณไม่อยากเรียนเลย?" showing 6 dimension chips. Student may tap any to exclude; tapped dimensions are zeroed in the final vector regardless of score. A "ไม่มี ข้ามได้เลย" option is prominently shown.

6. System applies L2-normalisation to the final 6-dimensional vector and stores it as the student's RIASEC profile. Student proceeds to recommendations via "ดูผลลัพธ์".

Alternative Flow — Targeted re-elicitation: If invoked from UC-02 with specific RIASEC dimensions flagged for clarification, the system skips Steps 1–2 and begins directly at Step 3, presenting only pairwise questions targeting the flagged dimensions. Steps 4–6 run normally. This path is triggered when the student selects "No" on the profile correction prompt in UC-02.
Postcondition: Student has a RIASEC-grounded interest profile used as input to the MVP program recommendation pipeline. Total elicitation time is 2–4 minutes across 12 steps. Career matching is Phase 2.

UC-02: View Program Recommendations
Actor: Guest / High School Student
Precondition: Interest profile (RIASEC vector) has been built via UC-01.
Main Flow:

1. System converts the student's RIASEC profile into a semantic interest query and structured weighting used for curriculum-side matching.

2. System runs MVP Pipeline B2: semantic search over indexed program, PLO, and course-description embeddings in Supabase pgvector.

3. System aggregates retrieved evidence by program and computes a curriculum-fit score from semantic similarity and available structured metadata.

4. Programs are ranked by curriculum-fit score. Top-N programs are selected for display.

5. System displays ranked program cards, each showing fit score, matched curriculum themes, and a plain-language explanation grounded in retrieved program evidence.

6. Student can tap any card to navigate to the full program detail page at `/programs/[program-id]` — a full-page view, not a modal.
7. System presents a profile correction prompt — “Do these recommendations feel right?” — on the first visit to the recommendation screen. If the student selects “No”, the system returns them to a targeted re-elicitation of the RIASEC dimensions most influential in the current ranking, preventing a poor initial vector from entering the interaction log.

Postcondition: Student sees a ranked list of KU programs with recommendations traceable from their interests to retrieved curriculum evidence. The full two-signal target architecture with O*NET and Neo4j is deferred to Phase 2.

UC-03: Explore PLO Spider Chart
Actor: Guest / High School Student
Precondition: A program card is selected.
Main Flow:

1. Student selects a KU program from the recommendation list.
2. System retrieves available PLO/profile data from Supabase.
3. System renders a static radar or profile chart showing the skill or outcome profile the program develops.
4. In Phase 2, the student’s interest profile may be overlaid on the chart for visual fit comparison.

Postcondition: Student has a visual understanding of the skills or outcomes a program builds. Interactive personal-fit overlay is outside MVP scope.

UC-04: Query KUru Advisor
Actor: Guest / High School Student
Precondition: มคอ.2 documents have been ingested and indexed in Supabase pgvector.
Main Flow:

1. Student enters a question in Thai or English in the chat interface.
2. System embeds the query locally using `intfloat/multilingual-e5-base`.
3. System retrieves the top semantically relevant มคอ.2 document chunks from pgvector.
4. System passes the chunks and question to Gemini 2.5 Flash with a strict citation instruction.
5. System displays the generated answer with inline มคอ.2 source citation badges.
6. The chatbot also handles TCAS questions for any program where admission data has been ingested — for example, "Computer Engineering Round 1 ต้องการ GPAX เท่าไหร่?" or "Architecture portfolio ต้องมีอะไรบ้าง?" Answers are grounded in the ingested TCAS data and include source citations. If TCAS data for a requested program has not been ingested, the system returns the standard fallback message.

Alternative Flow: If no relevant chunk is found, the system responds: "This information was not found in the curriculum document."
Postcondition: Student has received a cited answer grounded in the official curriculum document.

UC-05: View TCAS Admission Guide
Actor: Guest / Registered User
Precondition: TCAS admission data has been collected and structured per faculty and round. Students may also query TCAS information conversationally via KUru Advisor (UC-04), which draws from the same ingested data source.
Main Flow:

1. Student opens a program detail page or asks a TCAS-related question through KUru Advisor.

2. System displays or generates a grounded answer from available TCAS records, including applicable rounds, score requirements (GPAX, TGAT/TPAT/A-Level), portfolio criteria, fees, and deadlines where available.

3. If the requested TCAS field is missing from the ingested data, the system states that the information is not available rather than inferring an answer.

4. In Phase 2, students can navigate a standalone TCAS guide, filter by round, and compare faculties side by side.

Postcondition: Student understands the specific admission requirements available for the selected program or query, with missing information clearly identified.

UC-06: Explore Career Paths
MVP status: Phase 2 target-system use case; excluded from MVP pass/fail evaluation.
Actor: Guest / High School Student
Precondition: RIASEC interest profile has been built via UC-01.
Main Flow:

1. Student taps "ดูอาชีพที่เหมาะกับคุณ" (View matching careers) from the recommendation screen.

2. System retrieves the top 7 O*NET occupations (selected from the 10–15 matched internally during the recommendation pipeline) to display in the career explorer.

3. System displays a career list; each card shows the occupation title, a Thai-localised description, the top 3 required skills, and the KU programs that develop those skills.

4. Student taps a career card to expand detail.

5. System shows the full O*NET skill breakdown (35 standardized skills) and highlights which matched KU programs develop each skill, using Neo4j data.

6. Student can navigate directly from a career card to the relevant program card.

Postcondition: Student understands which careers align with their RIASEC interest profile and which KU programs prepare them for those careers.

UC-07: Refine Profile Through Implicit Signals
MVP status: Phase 2 target-system use case; excluded from MVP pass/fail evaluation.
Actor: Registered User
Precondition: Student is logged in and has an existing RIASEC profile from UC-01.
Main Flow:

1. Student browses program cards, saves programs, checks TCAS guides, and asks chatbot questions across one or more sessions.
2. System records behavioral signals alongside the PLO skill profiles of the associated programs, weighted by interaction type:
   - Program save — weight 1.0
   - TCAS guide viewed — weight 0.8
   - Chatbot query about a program — weight 0.6
   - Program card expanded — weight 0.4
   - Program card viewed — weight 0.1
3. System adjusts a blending weight α based on accumulated interaction data. A new user starts at $\alpha = 1.0$ (pure RIASEC-based recommendations). As interaction weight accumulates across 3–5 meaningful signals, α decays toward 0.2, shifting the ranking progressively toward behavioural signal. The transition is smooth and invisible to the student. At each session, the recommendation ranking is computed as:

   $$\text{score} = \alpha \times \text{RIASEC fit score} + (1 - \alpha) \times \text{behavioural fit score}$$

4. On the next session, the recommendation ranking reflects the updated blended profile.

5. Student can view a summary of how their recommendation ranking has shifted under "โปรไฟล์ของคุณ" on the saved profile dashboard, including a before/after comparison of the highest-ranked programs.

Alternative Flow: Student disagrees with the ranking shift and taps "รีเซ็ตโปรไฟล์" to clear accumulated interaction data and return to pure RIASEC recommendations ($\alpha = 1.0$).
Note on guest users: Interaction signals are only logged persistently for registered users. Guest interactions are tracked within the current session only and do not update α across visits. Guests always begin a new session at $\alpha = 1.0$ (pure RIASEC recommendations).
Postcondition: Recommendation ranking becomes progressively more personalised to the student's actual behaviour without requiring additional questionnaire steps. The underlying RIASEC vector from elicitation remains fixed; the behavioural fit score is derived from the individual student's own interaction history (not from other users) and operates as a separate signal blended at ranking time via α. True collaborative filtering — using patterns from students with similar profiles — is deferred to Phase 2.

UC-08: Explain Why a Program Was Recommended
Actor: Guest / High School Student
Precondition: Student is viewing a program recommendation card.
Main Flow:

1. Student taps "ทำไมถึงแนะนำคณะนี้?" (Why was this recommended?) on any program card.

2. System retrieves the MVP reasoning chain: student RIASEC profile $\rightarrow$ semantic curriculum query $\rightarrow$ retrieved program/PLO/course evidence $\rightarrow$ curriculum-fit score.

3. System generates a plain-language explanation via Gemini referencing the retrieved evidence: "Based on your interest

in [topic], this program was strongly matched because its curriculum includes [X] and [Y].”

4. In Phase 2, explanations may also include O*NET career matches and Neo4j PLO-skill paths.

Postcondition: Student understands the specific curriculum evidence behind a recommendation and can verify the cited source material.

UC-09: Search Programs Semantically
Actor: Guest / High School Student
Precondition: None — accessible without login.
Main Flow:

1. Student enters a natural language query in Thai or English in the semantic search bar on the PLO Explorer (e.g., “หลักสูตรที่เรียน AI เน้นโปรเจกต์ รับ TGAT ต่ำ”).
2. System embeds the query and optionally parses visible constraints such as topic areas, TCAS requirements, campus preferences, or teaching methods.
3. System executes semantic search against Supabase/pgvector and available structured metadata for indexed programs.
4. System returns ranked program cards matching the constraints, with an explanation of which constraints each program satisfies.
5. Student can refine the query or apply faceted filters (faculty, program level, TCAS round) where metadata is available.

Postcondition: Student finds programs matching specific criteria without browsing the full catalogue.

UC-10: Pin Programs and Compare
MVP status: Phase 2 target-system use case; excluded from MVP pass/fail evaluation.
Actor: Guest / High School Student

Precondition: At least 2 programs pinned via the pin button on any program card.
Main Flow:

1. Student pins programs from the explorer or recommendation screen. Pinned programs appear in a persistent tray at the top of the explorer (maximum 4).
2. Student taps "เปรียบเทียบ" from the pin tray.
3. System generates a side-by-side comparison: overlaid PLO radar charts, career path overlap, curriculum character profiles across 5 dimensions, TCAS accessibility per round, and personalised fit scores relative to the student's RIASEC profile.
4. Student can tap "เช็ค Portfolio ทั้งหมด" to run a portfolio gap analysis against all pinned programs simultaneously, showing which program the student is most ready for right now.
5. Student can tap "แชทกับ KUru" to open the chatbot with all pinned programs as active context.

Postcondition: Student has a clear comparative picture of their shortlisted programs and knows which they are most ready to apply for.

UC-11: View Curriculum Timeline
MVP status: Phase 2 target-system use case; excluded from MVP pass/fail evaluation.
Actor: Guest / High School Student
Precondition: Student is on a program detail page.
Main Flow:

1. Student selects the "ชีวิตนักศึกษา" tab on a program detail page.
2. System retrieves course structure data extracted from มคอ.2 during ingestion: courses per year, credit hour distribution, teaching method breakdown (lecture/lab/project ratios), assessment method types, and sample learning activity descriptions.

3. System displays a year selector (ปี 1–4). Default shows Year 1.

4. For the selected year, system displays a Gemini-generated narrative of what that year involves, course composition breakdown, workload indicator, and one concrete example activity description extracted from มคอ.2.

5. Student taps different years to compare progression across the program.

Postcondition: Student has a concrete sense of what studying this program year by year actually involves before applying.

UC-12: Portfolio Readiness Check
MVP status: Phase 2 target-system use case; excluded from MVP pass/fail evaluation.
Actor: Guest / High School Student
Precondition: Student has a portfolio PDF. Target program has portfolio criteria data ingested.
Main Flow:

1. Student navigates to the Portfolio Coach from the nav bar, a program detail page, or the pin tray.

2. Student selects target program(s) from the supported Phase 2 portfolio-checking set and uploads their portfolio PDF.

3. System uses Gemini to extract a structured portfolio profile: activities classified by type and level, GPAX, certificates, personal statement presence.

4. System retrieves the pre-extracted faculty criteria schema for the selected program(s).

5. System performs a four-part gap analysis: (1) hard threshold eligibility check; (2) required item coverage; (3) preferred item strength assessment with level-aware scoring;

(4) qualitative criteria assessment using multi-step Gemini reasoning.

6. System displays a structured report: eligibility status, portfolio strength profile, prioritised gap list, and deadline-aware action recommendations.
7. If multiple programs are selected, system shows which program the student is most ready for right now.

Postcondition: Student knows exactly what is missing from their portfolio and what to prioritise before the application deadline. The system provides preparation guidance, not admission probability prediction.

3.3 Use Case Diagram

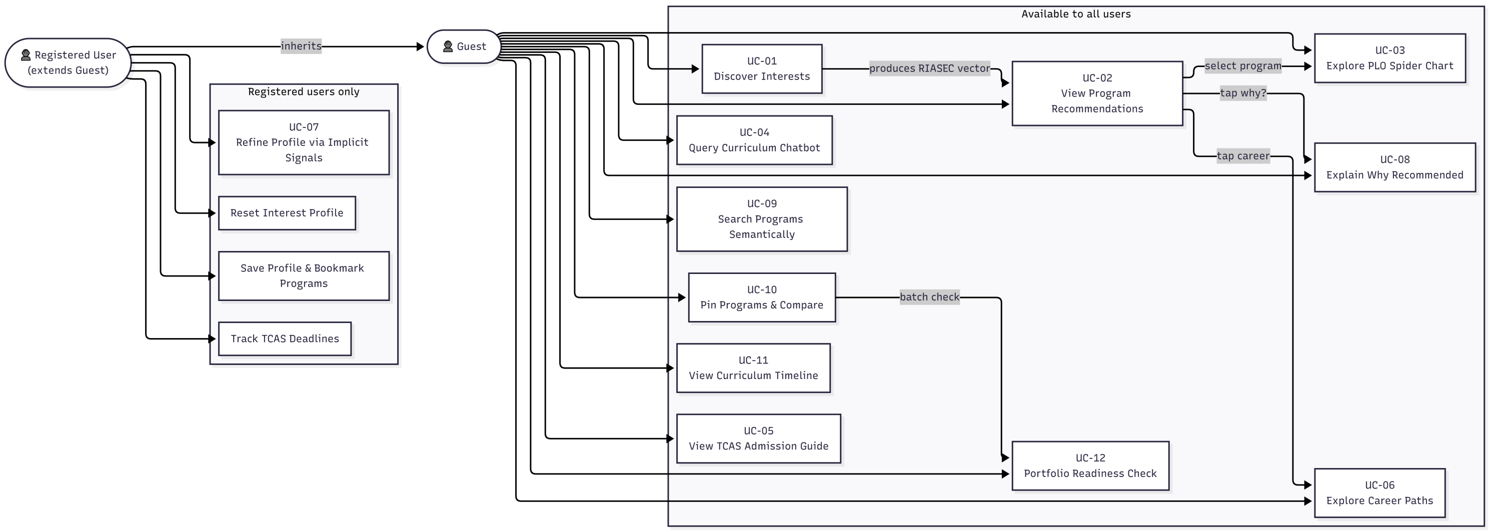

Figure 3.1: KUru Use Case Diagram

3.4 Non-Functional Requirements

- **Performance:** The RAG chatbot must return a grounded response within 5 seconds for 95% of queries under normal load. The recommendation pipeline must return ranked results within 8 seconds of interest profile submission. The PLO spider chart must render within 2 seconds of program selection.

- **Availability:** The web application must maintain at least 95% uptime during the TCAS application period (January–April). Planned maintenance must be performed outside this window.

- **Scalability:** The system must support at least 100 concurrent users during peak TCAS periods without degradation of the above response time targets. Supabase connection pooling and Vercel's serverless deployment model are relied upon to meet this target without custom infrastructure.

- **Security:** All user authentication is delegated to Supabase Auth (OAuth 2.0 / magic link) where login-based features are enabled. Portfolio PDF processing is Phase 2; when implemented, uploaded documents are processed ephemerally via the Gemini API and are not retained on the server after processing. API keys for Gemini and Neo4j are stored as environment variables and never exposed to the client.

- **Data Privacy (PDPA):** Interest profiles and any future portfolio PDFs contain personal data. In accordance with Thailand's Personal Data Protection Act B.E. 2562 (PDPA), users are notified of data usage when profile storage or upload features are used. Guest interaction data is session-scoped and not persisted. Registered users may delete their stored profile and interaction history at any time via the dashboard once Phase 2 persistence is enabled.

- **Accessibility:** The interface must meet WCAG 2.1 Level AA contrast requirements. All interactive elements must have accessible labels for screen reader compatibility. Thai and English language switching must be available on every screen.

- **Maintainability:** TCAS admission data and มคอ.2 curriculum documents must be re-ingestible independently of the application codebase. The data ingestion pipeline must produce a full re-index without manual code changes when new PDF documents are supplied.

3.5 User Interface Design

The interface follows a mobile-first web design using Next.js, Tailwind CSS, and Shadcn/UI. The key screens are described below.

Landing Page. The entry point presents a bilingual Thai/English value proposition with two primary call-to-action paths: *ค้นหาความสนใจ* (interest discovery) for undecided students, and *ดูโปรแกรม* 70+ คณะ มก. (browse programs) for students who already have a direction. Supporting copy highlights the three-minute completion time, มคอ.2 data coverage, and free access.

Interest Discovery. The twelve-step adaptive elicitation flow collects the student's RIASEC profile. Steps 1–6 present one Likert screen per RIASEC dimension (4 Thai-language statements, 5-point scale, colour-coded per dimension); Step 7 applies a global confidence scalar; Step 8 shows adaptive pairwise forced-choice questions for ambiguous dimension pairs (maximum 4–6, threshold delta < 3); Steps 9–11 present three scenario questions with A–F role options; Step 12 displays the profile summary with Holland Code, per-dimension score bars, and optional deal-breaker filter chips. A progress bar is shown throughout.

Figure 3.2: KUru landing page — bilingual entry point with interest-discovery and program-browse paths.

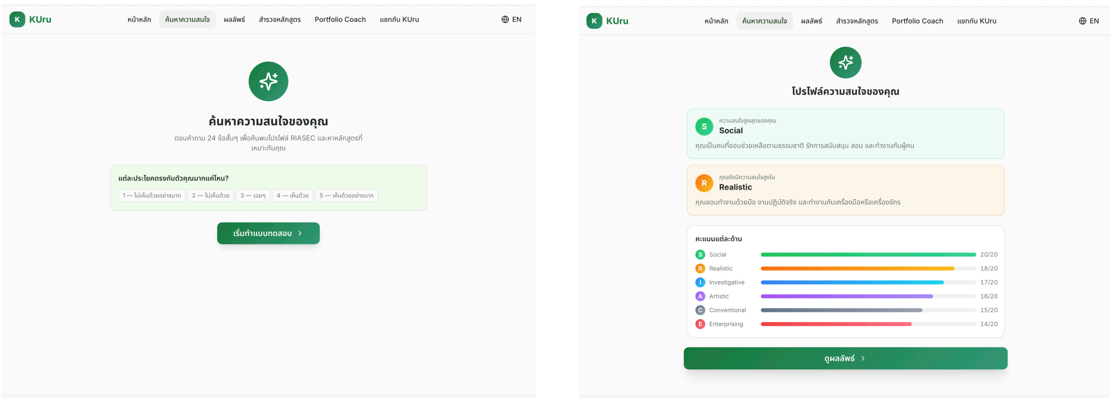

Figure 3.3: Interest discovery flow: (left) opening screen prompting the student to begin the 24-statement RIASEC assessment; (right) Step 12 profile summary showing the Holland Code, dominant dimensions (Social, Realistic), and per-dimension score bars before proceeding to recommendations.

Recommendation Results (ผลพัจน์). After elicitation, the MVP presents a personalised results page structured in two sections: (1) the student's RIASEC profile card displaying their Holland Code and top dimensions; and (2) ranked KU program cards with curriculum-fit scores derived from Pipeline B2 semantic matching. Each card links the fit score to retrieved curriculum evidence. On first visit, a profile correction prompt allows the student to trigger targeted re-elicitation if the ranking does not feel right. O*NET career cards and the full hybrid Pipeline A+B ranking are Phase 2.

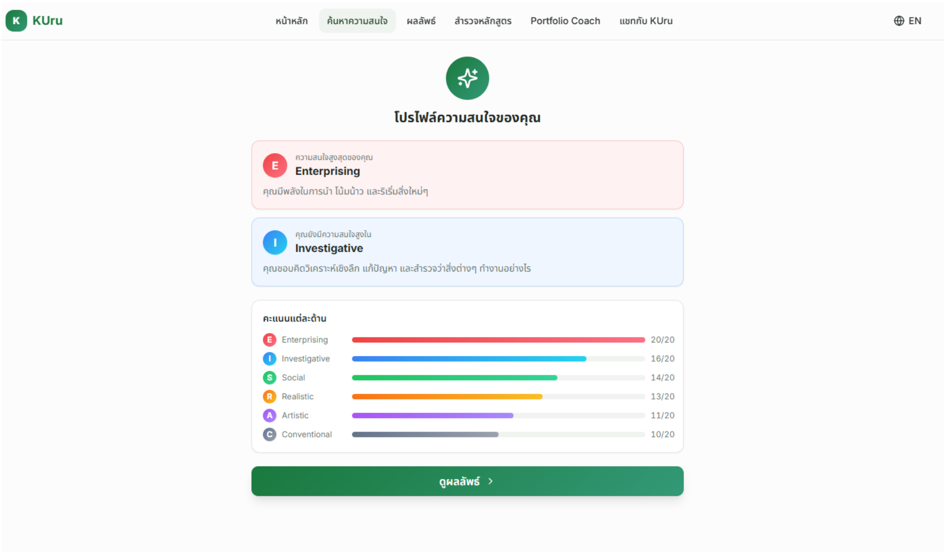

Figure 3.4: Recommendation results page — MVP shows the RIASEC profile summary and ranked curriculum-evidence program cards; O*NET career cards shown in the mockup are Phase 2.

Program Explorer (สำรวจหลักสูตร). A browsable and semantically searchable directory of KU programs. In the current POC and evaluated MVP, each program card displays available program metadata, faculty/campus tags, curriculum/PLO evidence summaries, and available TCAS or fee fields. Students can filter by faculty or keyword and search using natural-language Thai or English queries. Pin-tray comparison is Phase 2.

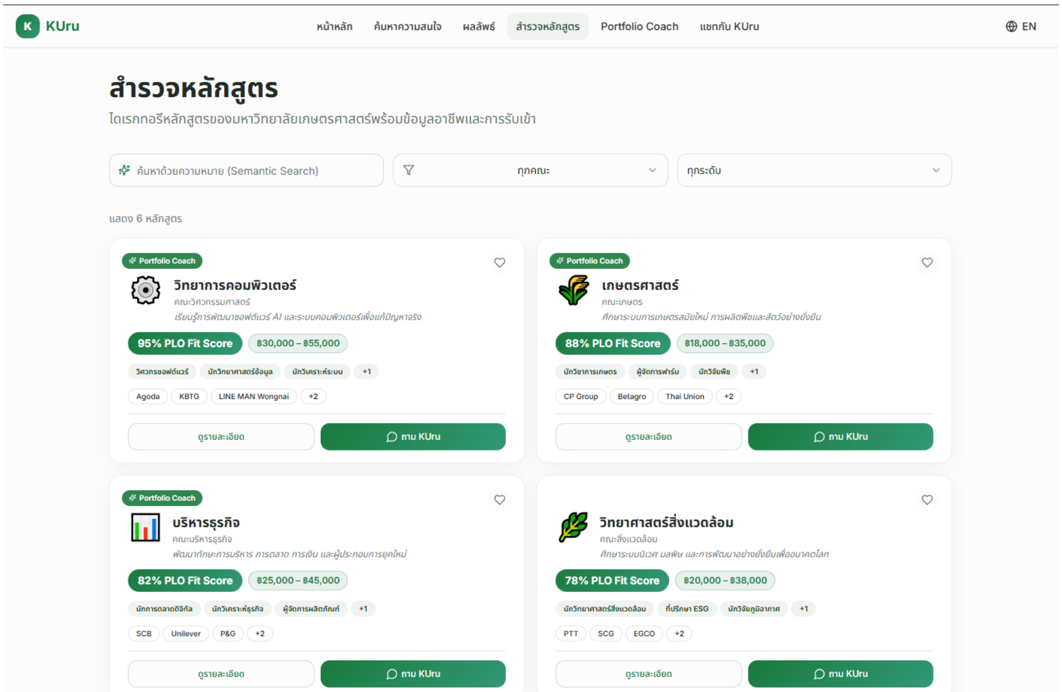

Figure 3.5: Program explorer — grid of program cards with available curriculum, faculty, and TCAS metadata. The search bar supports natural language queries in Thai and English.

KUru Advisor (แชทกับ KUru). A RAG-powered conversational interface for querying curriculum and TCAS information. The chatbot answers natural language questions in Thai and English, grounding responses in ingested มคอ.2 documents and TCAS admission data. Each response includes source citations (document name and page). Multi-program pinned context is Phase 2.

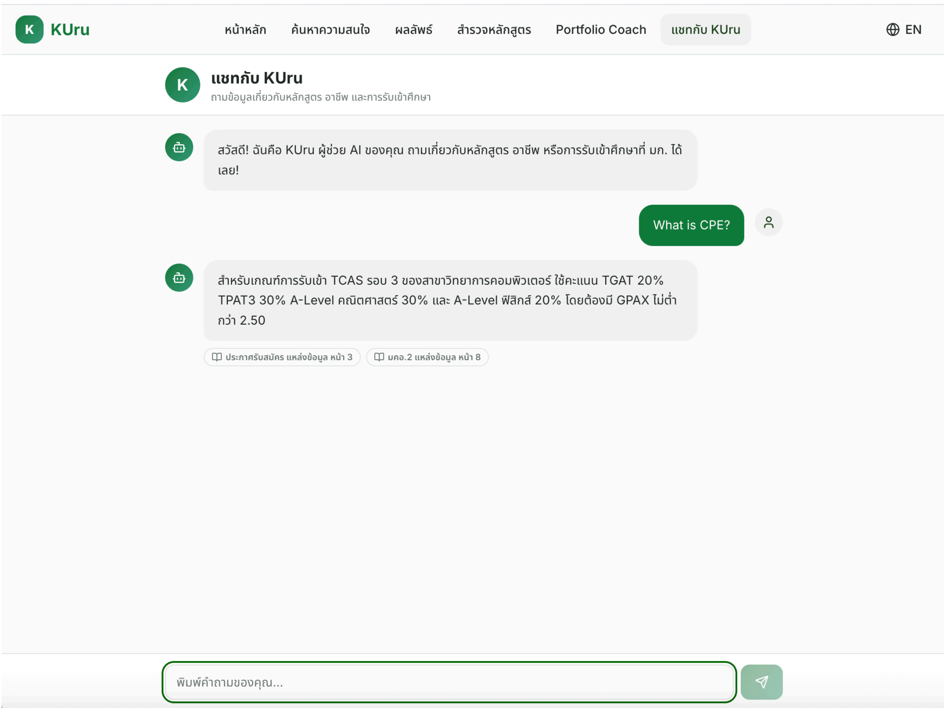

Figure 3.6: KUru Advisor chat interface — the student asks “What is CPE?” and receives a grounded answer about Computer Engineering TCAS Round 3 requirements with source citations.

Portfolio Coach (Phase 2). A two-step portfolio readiness checker accessible from the navigation bar, program detail page, or pin tray. In the first step the student uploads portfolio documents (PDF or image). In the second step the system displays a readiness score, a gap analysis comparing the student’s credentials against the target program’s admission criteria, and a checklist of fulfilled and missing items.

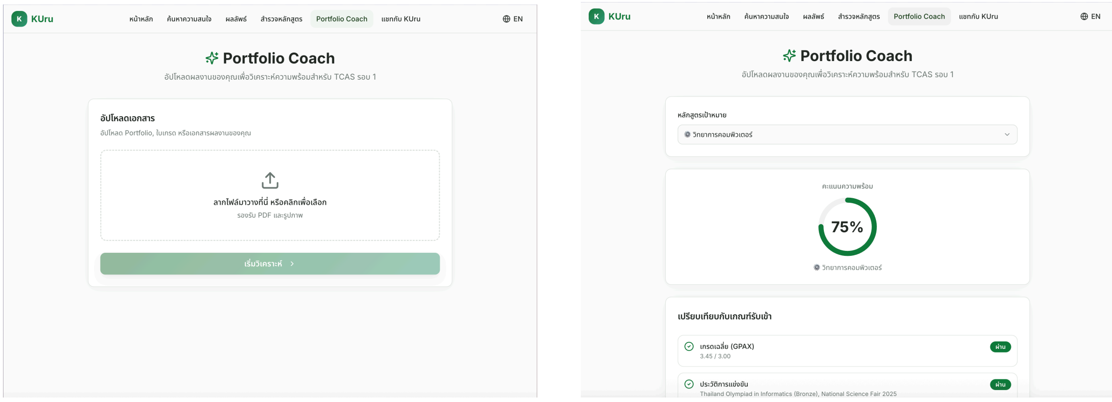

Figure 3.7: Portfolio Coach Phase 2 mockup: (left) document upload screen accepting portfolio PDFs and transcript images; (right) analysis results showing a readiness score and criteria checklist.

Program Detail Page. Full-page view at `/programs/[program-id]` — not a modal — to support URL sharing, extended reading, and mobile scrolling. The MVP page contains program identity, curriculum/PLO evidence, available TCAS information, source-linked chatbot context, and a static PLO/profile chart where extracted data is available. Career sections, curriculum timeline tabs, personal eligibility checking, portfolio checker entry points, and PLO overlay are Phase 2.

Program Comparison View (Phase 2). Side-by-side comparison triggered from the pin tray (UC-10). Shows overlaid PLO radar charts, career path overlap, curriculum character profile across five dimensions, TCAS accessibility per round, and RIASEC fit scores for up to four pinned programs simultaneously.

Career Explorer (Phase 2). List of O*NET occupations matched to the student's RIASEC profile via Pipeline A. Each card shows the Thai-localised occupation title, required skill clusters, and the KU programs that develop those skills.

TCAS Guide. Faculty-specific breakdown of applicable admission rounds, score requirements (GPAX, TGAT, TPAT, A-Level), portfolio criteria, and application deadlines, drawn from the ingested TCAS dataset.

Saved Profile Dashboard (Phase 2). Bookmarked programs, TCAS deadline tracker, interest profile summary, and a before/after comparison of recommendation ranking shifts over time. Requires student login.

Chapter 4

Software Architecture Design

4.1 System Architecture Overview

KUru is a three-tier web application with a distinct AI layer. The frontend (Next.js) communicates with a Python FastAPI backend, which orchestrates grounded retrieval, recommendation, and structured program-data services. The architecture is described in two layers: the evaluated MVP architecture and the target-system architecture.

4.1.1 Evaluated MVP Architecture

The evaluated MVP uses Supabase PostgreSQL with pgvector as the primary data layer. The backend supports: (1) a structured RAG pipeline for curriculum, TCAS, and fee Q&A; (2) RIASEC interest-profile storage and scoring; (3) simplified Pipeline B2 recommendation based on semantic matching against indexed program, PLO, and course content; (4) semantic program search; and (5) static PLO/profile visualisation. Gemini 2.5 Flash Lite via OpenRouter is used for grounded answer generation and explanation, while local multilingual E5 embeddings are used for Thai-English retrieval.

Neo4j, O*NET career matching, behavioural re-ranking, portfolio checking, program comparison, and curriculum timeline generation are not required for MVP pass/fail evaluation. They remain part of the target-system design.

4.1.2 Target-System Architecture

The target system extends the MVP with a Neo4j knowledge graph connecting faculties, PLOs, skill clusters, and careers; O*NET occupation matching; behavioural personalisation; portfolio readiness analysis; curriculum timeline extraction; and multi-program comparison. In that target architecture, both RAG and recommendation read from a shared data layer consisting of Supabase (PostgreSQL + pgvector for embeddings) and Neo4j (knowledge graph on Railway).

Figure 4.2 shows the target-system domain model: the eleven core entities and their relationships. The MVP implements the subset centred on **Student**, **RIASECProfile**, **Program**, **Faculty**, **PLO**, **TCASRecord**, and **Course** stored in Supabase. Phase 2 adds persistent **InteractionLog**, **PortfolioCriteria**, **SkillCluster**, and **Career** relationships through Neo4j. In the target system, PLOs are linked to **SkillCluster** nodes in Neo4j (the *develops* relationship), and **Career** nodes are linked to the same clusters (the *requires* relationship) — this shared vocabulary enables the multi-hop recommendation path from student interests to program outcomes.

4.2 Software Design

4.2.1 RAG Pipeline Sequence

When a user submits a query to the curriculum chatbot: (1) the query is embedded locally using `intfloat/multilingual-e5-base`; (2) the embedding is used to perform a cosine similarity search against the pgvector store of มคอ.2 document chunks; (3) the top-k retrieved chunks are assembled into a prompt context along with the user query and the conversation history; (4) Gemini 2.5 Flash Lite via OpenRouter generates a grounded response in the requested language; (5) source citations are appended to the response. In the current POC, citation chips expose provenance such as source file, section type, and similarity/confidence context, but they do not yet open the exact PDF page or extracted chunk. Deep source inspection is Phase 2; the future API should include `chunk_id`, page number, and source URL so the frontend can open a source preview panel or PDF page.

4.2.2 Recommendation Pipeline Sequence

For the evaluated MVP, recommendation uses the simplified Pipeline B2 path. RIASEC is not treated as proof that a student will be satisfied with a program after graduation. Instead, it is used as an interest signal that can be converted into curriculum search intent. When a student completes the interest discovery flow, the backend converts the RIASEC profile into interest themes and a semantic query, retrieves semantically similar program/PLO/course chunks from Supabase pgvector, aggregates evidence by program, computes a curriculum-fit score, and generates a cited plain-language explanation from the retrieved evidence. This is the recommendation behaviour evaluated for MVP pass/fail.

For example, a high Investigative–Realistic profile is converted into themes such as analysis, problem solving, systems, technical design, experiments, and hands-on laboratory work. These themes are embedded as a semantic query and matched against curriculum evidence such as PLOs, course descriptions, project-based learning statements, laboratory requirements, and technical skill outcomes. A program ranks highly only when retrieved curriculum chunks support the match.

The full target-system recommendation pipeline proceeds in five stages using two independent parallel signals:

1. **RIASEC vector construction.** The twelve-step adaptive elicitation (UC-01) produces a weighted 6-dimensional RIASEC vector. Steps 1–6 (Likert screens) produce raw scores per dimension (max 20 each). Step 7 applies a global confidence scalar (1.0, 0.75, or 0.5). Step 8 (adaptive pairwise) adjusts scores for dimension pairs with delta < 3. Steps 9–11 (scenario questions) adjust scores proportionally per response. Step 12 applies the optional dealbreaker filter, zeroing out explicitly rejected dimensions. The resulting vector is L2-normalised to a unit vector. The MVP uses this vector to form a Pipeline B2 curriculum search intent; the target system also uses it in Pipeline A and Neo4j Pipeline B1.

2. **Parallel signal computation.** Two independent pipelines run concurrently:

   Pipeline A (Career-side): The student RIASEC vector is compared against precomputed O*NET occupation RIASEC interest profiles in Supabase pgvector using cosine similarity. O*NET interest profiles are empirically derived from surveys of employed workers, making this step grounded in real-world data. The top 10–15 occupations form the internal career space; the top 7 are surfaced to the student in the Career Explorer (UC-06). An A-score per KU program is computed based on career-program pathway alignment. O*NET's 35-skill taxonomy is used for Career Explorer display in UC-06 only — it is not used as a bridge to Neo4j in the ranking pipeline.

   Pipeline B (Curriculum-side): Two sub-signals computed independently and combined:

   - **B1 — Neo4j PLO match:** The student's RIASEC vector is converted to a SkillCluster weight vector and matched against KU program PLO SkillCluster profiles via Neo4j graph query. B1-score is the cosine similarity between the student weight vector and each program's PLO skill profile.
   - **B2 — Semantic course content match:** The student's interest query is matched against course description embeddings in Supabase pgvector via semantic search over ingested มคอ.2 course content. B2-score is the weighted mean similarity of the top retrieved course chunks per program.

   Combined B-score $= 0.4 \times$ B1 $+ 0.6 \times$ B2.

   The two pipelines are independent — an error in Pipeline A does not propagate into Pipeline B.

3. **Score synthesis.** Programs are ranked by a final score combining both pipeline outputs:

   $$\text{score} = 0.35 \times \text{A-score} + 0.65 \times \text{B-score}$$

   The top-N programs are selected for display.

4. **Explanation generation.** The top-N ranked programs, together with their Pipeline A details (matched O*NET occupations) and Pipeline B details (curriculum themes and matched PLOs), are passed to Gemini 2.5 Flash as structured context. Gemini generates a plain-language explanation for each program referencing both career alignment and curriculum fit. Explanations are generated in Thai or English depending on the student's language setting.

5. **Behavioural re-ranking (registered users only).** For registered users with accumulated interaction data, the pipeline scores from Stage 3 are blended with a behavioural fit score derived from the individual student's own interaction history (not from other users — true collaborative filtering across students is deferred to Phase 2). The blend ratio is controlled by α: new users start at $\alpha = 1.0$ (pure pipeline score); α decays toward 0.2 as interaction weight accumulates according to the threshold schedule documented in UC-07 (3–5 meaningful signals $\rightarrow \alpha \approx 0.5$; regular user $\rightarrow \alpha \approx 0.2$). The final ranking score is:

   $$\text{final score} = \alpha \times \text{pipeline score} + (1 - \alpha) \times \text{behavioural fit}$$

   Guest interaction signals are session-scoped and do not update α; guests always begin a new session at $\alpha = 1.0$. Fallback hierarchy when behavioural data is sparse: (1) sufficient interaction history $\rightarrow$ normal blended signal; (2) limited interaction history $\rightarrow$ α held higher; (3) no interaction history $\rightarrow$ pure pipeline score ($\alpha = 1.0$). Interaction weights are documented in UC-07.

4.2.3 Interest Elicitation Flow

The interest elicitation flow (UC-01) produces the RIASEC vector that seeds the MVP recommendation pipeline. In Phase 2, the same vector can also seed the Career Explorer. The flow is adaptive: the pairwise comparison step (Step 8) is shown only when at least one dimension pair has a score delta below the 3-point ambiguity threshold; students with a clearly differentiated Likert profile skip directly to the scenario questions. Figure 4.6 shows the full conditional logic from entry to RIASEC vector output.

4.3 AI and Data Design

4.3.1 Data Sources and Collection

- **KU curriculum documents (มคอ.2)**: Provided directly by KU faculty advisors for all KU programs. Thai-language PDFs of varying length (typically 20–80 pages per program). These are the primary knowledge base for both the RAG pipeline and the PLO extraction pipeline. Documents are version-tagged by academic year and re-ingested when updated.
- **O*NET occupational database [7]:** Publicly available from the US Department of Labor. Contains two data layers used by KUru: (a) *Interests* — the RIASEC profile for each occupation, derived from surveys of employed workers, used for career matching from the student's RIASEC vector; and (b) *Skills* — 35 standardized skills per occupation with importance and level scores, displayed in the Career Explorer (UC-06) to show students what skills their matched careers require. Not used as a bridge to Neo4j in the ranking pipeline — Pipeline B1 queries Neo4j directly from the RIASEC-derived SkillCluster weight vector. Downloaded as structured data (CSV/JSON). Occupation titles are mapped to Thai equivalents for student-facing

display; skill and interest data are used directly as the underlying structure is language-independent.

- **TCAS admission data:** Ingested from three sources in priority order. (1) *KU faculty-provided project PDFs* received via Google Drive — each PDF (e.g., โครงการช้างเผือก.pdf) may contain admission records for multiple programs across multiple faculties. A batch downloader retrieves all PDFs recursively. Each PDF is classified as tcas / mko2 / portfolio / unknown using rule-based keyword detection on the first page. TCAS PDFs are processed by a structured extraction pipeline using Gemini to parse per-program TCASRecord objects covering project name, round, faculty, program name, quota, GPAX minimum, exam score criteria, portfolio requirements, and application deadlines. (2) *mytcas.com* — scraped as a gap-filling layer for programs not covered by faculty PDFs, using the mytcas_code field in the canonical program registry. (3) *Faculty portfolio criteria PDFs* — retained as a Phase 2 source for the Portfolio Readiness Checker rather than an MVP pass/fail requirement. Available sources are resolved against a canonical program registry using fuzzy name matching with PyThaiNLP normalization and sequence similarity scoring (threshold 0.85). Records below this threshold are flagged for manual review. A Supabase materialized view (`programs_unified`) joins มคอ.2 structured data, TCAS records, and available admission metadata into a unified program record keyed on program_id, refreshed after each ingestion run.

- **Interaction log (Phase 2):** Generated by the system as registered users engage with KUru. Stored in Supabase PostgreSQL. Records program saves, card expansions, TCAS guide views, chatbot queries, and card views, each with a timestamp, interaction weight, and associated program identifier. Used to compute the behavioural fit score in

Stage 5 of the target-system recommendation pipeline. Not required for MVP pass/fail evaluation.

- **Faculty portfolio criteria documents (Phase 2):** Published by KU faculties for TCAS Round 1 and Round 2 applications. Ingested via the same PDF extraction pipeline as มคอ.2 using Gemini in structured extraction mode to produce a per-faculty, per-round criteria schema. Stored in Supabase PostgreSQL alongside TCAS admission data. Updated at the start of each TCAS cycle. Used by the Portfolio Readiness Checker.

- **Course structure data (partial MVP / Phase 2):** Extracted from มคอ.2 during ingestion as a structured layer stored separately from the embedding chunks where available. MVP uses available course and PLO text for RAG, semantic search, and simplified recommendation. Full per-year course timeline visualisation is Phase 2.

มคอ.2 document ingestion uses a six-stage pipeline:

1. **PDF classification.** Pages are classified by text density and Thai character density (minimum 20% Thai Unicode characters for Thai-language documents, minimum 100 characters per page) into born-digital, scanned, pure-image, or mixed types.

2. **Tiered text extraction.** The current curriculum ingestion pipeline first runs PyMuPDF on every PDF page. Pages with very low text yield are rendered to images and sent to Typhoon OCR (`typhoon-ocr`) when `TYPHOON_API_KEY` is available; recent targeted re-ingests mostly record `pymupdf+typhoon_pages`. If the whole PDF remains below the scanned-document threshold, the ingest script treats it as a full scanned PDF and calls the isolated `ocr_extractor.extract_with_ocr()` path. That full scanned-PDF path uses the configured `OCR_MODEL`:

by default direct Gemini `gemini-2.5-flash`, or Typhoon/OpenRouter if explicitly configured, with Tesseract as a last local fallback when vision OCR yields too little text. Each page is tagged with its extraction method (`pymupdf`, `typhoon_page`, `scanned`, or `gemini_ocr`) for coverage reporting.

3. **Section-aware chunking.** PLO sections, course descriptions, admission requirements, and curriculum structure sections are chunked separately (target 500 tokens, 50-token overlap) to preserve semantic coherence. Each chunk is tagged with section type (`plo` / `course` / `admission` / `curriculum_structure` / `general`).

4. **Structured extraction.** After text extraction and chunking, Gemini text mode (`LLM_MODEL=gemini-2.5-flash-lite` by default) extracts program metadata, PLOs, course lists, year timeline fields, and curriculum mapping into structured JSON. The MVP stores these fields in Supabase for program detail pages, semantic search, RAG grounding, and coverage reporting. Neo4j population from PLO/skill data is Phase 2.

5. **Embedding generation.** The current POC uses the local multilingual E5 embedding model (`intfloat/multilingual-e5-base`) to embed chunks before upserting them to Supabase/pgvector. This avoids external embedding quota limits. Chunks from failed OCR pages are excluded.

6. **Target-system graph population and quality reporting.** For Phase 2, Program → PLO → SkillCluster nodes and edges are created in Neo4j from the structured PLO extraction output. For the MVP, the corresponding quality report focuses on whether curriculum, TCAS, fee, and PLO chunks are retrievable from Supabase/pgvector and whether

documents with OCR or extraction failures are flagged for manual review.

4.3.2 Model Design

The primary answer-generation model for the MVP RAG path is Gemini 2.5 Flash Lite via OpenRouter, selected for Thai/English response quality and cost efficiency. Direct Gemini text-mode calls are used for structured extraction where appropriate, and direct Gemini vision OCR is used only in the isolated full scanned-PDF OCR path when configured. The project avoids LangChain or similar orchestration frameworks to reduce latency and maintain explicit control over prompt structure.

Embeddings use the local `intfloat/multilingual-e5-base` model, with the query/passage prefixes required by the E5 family. This model produces multilingual semantic representations for Thai and English while avoiding Gemini embedding quota limits.

The Phase 2 Neo4j knowledge graph uses a schema with four node types: Faculty, PLO, SkillCluster, and Career. Edges represent: Faculty -[HAS_PLO]→ PLO, PLO -[DEVELOPS]→ SkillCluster, Career -[REQUIRES]→ SkillCluster. This schema supports the multi-hop queries required for the target-system recommendation: given a student's SkillCluster profile, find Faculties whose PLOs develop those clusters.

The interest discovery interface uses a 12-step elicitation flow grounded in Holland's RIASEC model [1]. Steps 1–6 present one RIASEC dimension per screen with 4 Likert statements each (5-point scale, max score 20 per dimension). Step 7 applies a global confidence scalar (1.0 / 0.75 / 0.5) to all dimension scores based on a single self-reported confidence question. Step 8 presents adaptive pairwise forced-choice questions targeting only dimension pairs where the score delta is below a 3-point threshold, with a maximum of 4–6 pairs shown. Steps 9–11 present three scenario questions with A–F role options mapped to RIASEC dimensions, adjusting scores proportionally. Step 12 displays the profile summary with an optional dealbreaker filter that zeros out any explicitly rejected dimensions before L2-normalisation. The full question bank is documented in Appendix A.

The MVP recommendation ranking uses Pipeline B2 curriculum-side semantic fit only. The Phase 2 target ranking blends the RIASEC-derived fit score with a behavioural fit score using a blending weight α that decays automatically as the student accumulates interaction data: new users start at $\alpha = 1.0$ (pure RIASEC); after 3–5 meaningful interactions $\alpha \approx 0.5$ (balanced blend); regular users converge toward $\alpha \approx 0.2$ (behaviour-dominated). Interaction signals are weighted by type: program save (1.0), TCAS guide viewed (0.8), chatbot query about a program (0.6), card expanded (0.4), card viewed (0.1).

The baseline for comparison is keyword-only `ilike` search over the same มคอ.2 corpus without embedding-based retrieval or LLM generation. The full RIASEC-to-topic mapping is documented in Appendix A.

4.3.3 Evaluation

- **Current POC baseline (May 2026):** The current RAG POC is evaluated with custom LLM-as-judge regression tests tracked in MLflow on a 0–3 scale, including TCAS, fee, alias, bilingual, and missing-data honesty checks. The selected production regression run, `v8_structured_tcas_fees`, records 72.7% good answers and an average score of 2.055/3.0 on the fee/TCAS regression suite. The headline retrieval benchmark, `latest_submission_headline_v7_rerank_74pct`, records 74% good answers and an average score of 2.26/3.0. RAGAS was considered but is not the primary POC metric because custom checks better capture Thai/English KU-specific correctness, structured TCAS/fee evidence, and missing-data honesty. This baseline is reported honestly as POC evidence, not as final MVP success.
- **Final MVP RAG quality:** RAGAS framework metrics [3] are added as standardized supplementary metrics for UC-04 and UC-08: Faithfulness (proportion of claims in the answer supported by retrieved context), Answer Relevancy (semantic similarity between answer and question),

and Context Recall (proportion of relevant information retrieved). Target: Faithfulness > 0.80, Answer Relevancy > 0.75. Custom regression tests remain required for TCAS, fees, aliases, bilingual behaviour, and missing-data honesty because these behaviours are KU-specific.

- **Recommendation quality:** The MVP does not claim to measure whether a student will remain satisfied with a selected program after four years. That outcome is longitudinal and cannot be observed within the project period. Instead, recommendation quality is evaluated as a short-term relevance and explainability task: given a manually curated set of 30–50 student interest profiles, the system must rank programs judged relevant by faculty advisors, senior students, or documented curriculum evidence. Metrics are Mean Reciprocal Rank (MRR) and NDCG@5, with target MRR > 0.60. Explanations are also checked for faithfulness to the retrieved curriculum evidence. The MVP evaluates the simplified Pipeline B2 recommender; career-path validation, full hybrid Pipeline A+B, and Neo4j B1 evaluation are Phase 2.

- **Interest discovery and semantic search usability:** UC-01 completion rate target $> 80\%$, with System Usability Scale (SUS) score from user testing with 10–15 high school students or KU first-year students. UC-09 is evaluated through semantic query checks and user task completion. Target: SUS score > 70 (acceptable).

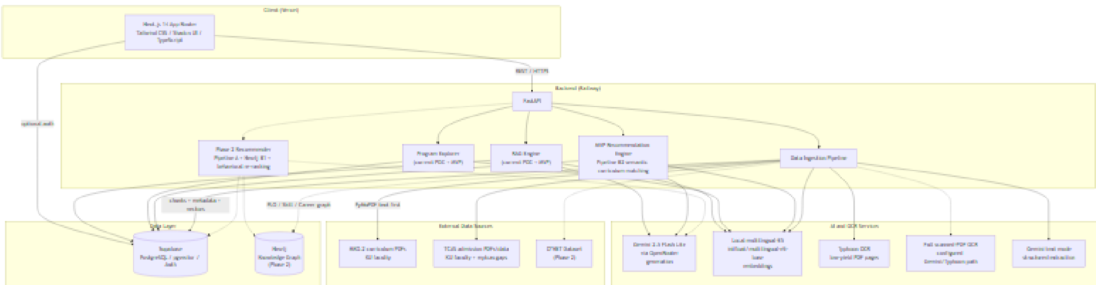

Figure 4.1: KUru System Architecture

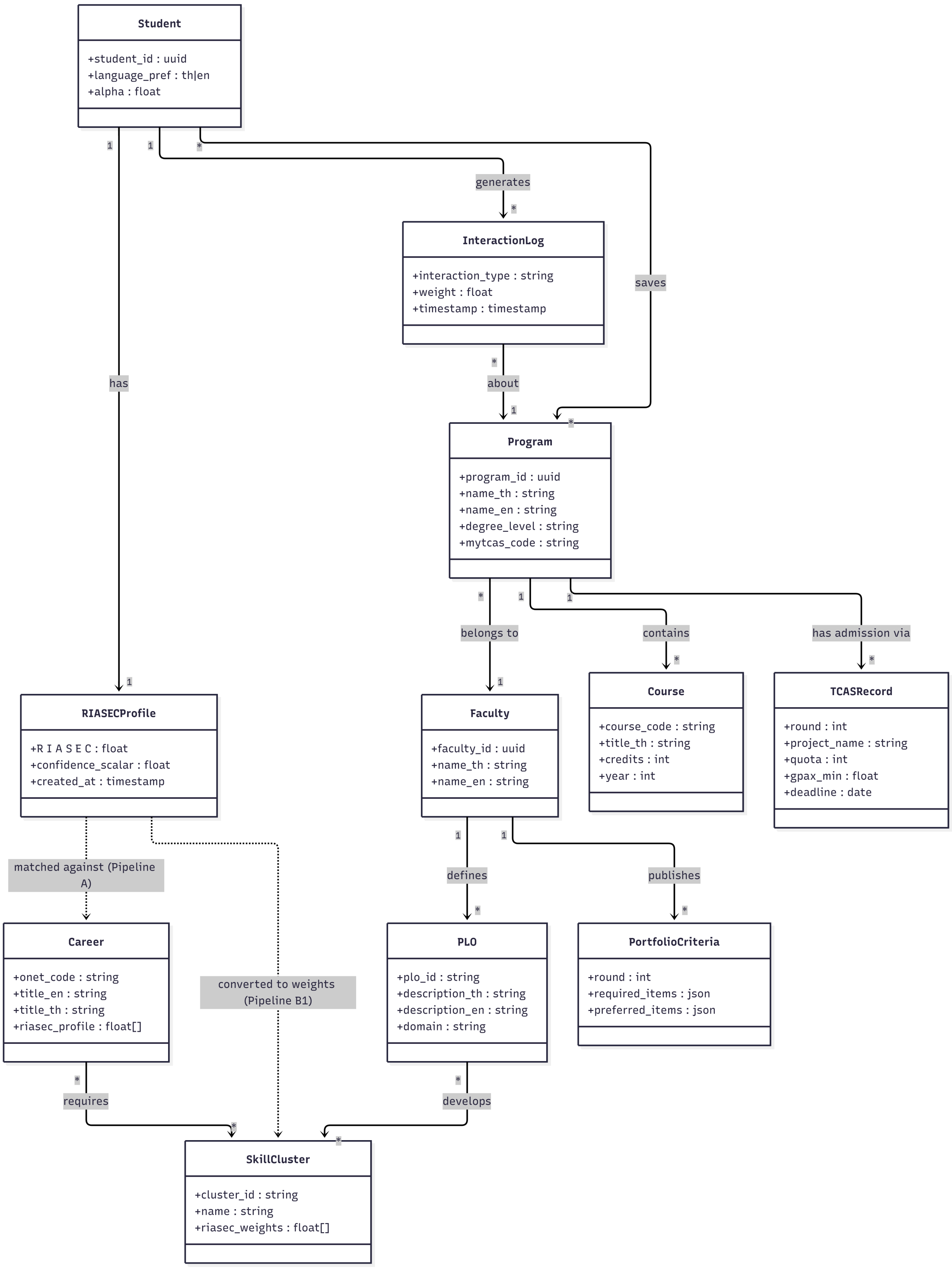

Figure 4.2: KUru Domain Model

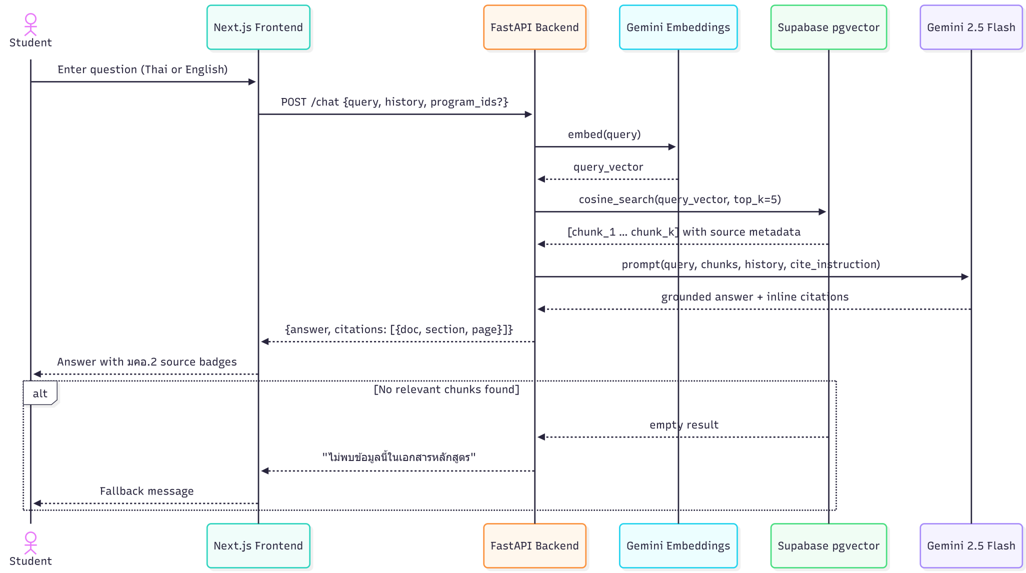

Figure 4.3: RAG Pipeline Sequence

MVP recommendation flow: RIASEC answers → 6D RIASEC profile → top dimensions / Holland code → interest themes and semantic query → Supabase/pgvector search over program, PLO, and course chunks → group evidence by program → curriculum-fit score → ranked programs with evidence-grounded explanation.

Figure 4.4: MVP Recommendation Pipeline: RIASEC-to-Curriculum Semantic Matching

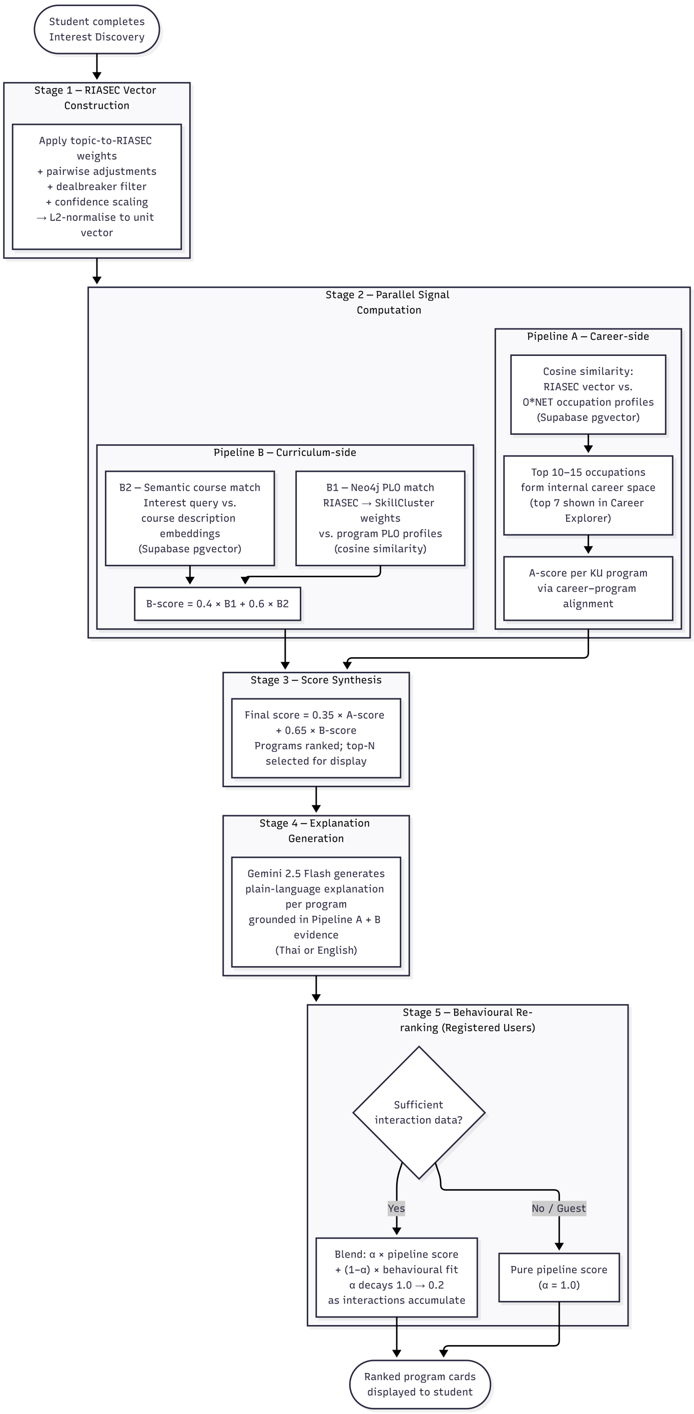

Figure 4.5: Phase 2 Target-System Recommendation Pipeline — Five Stages with Two Parallel Signals

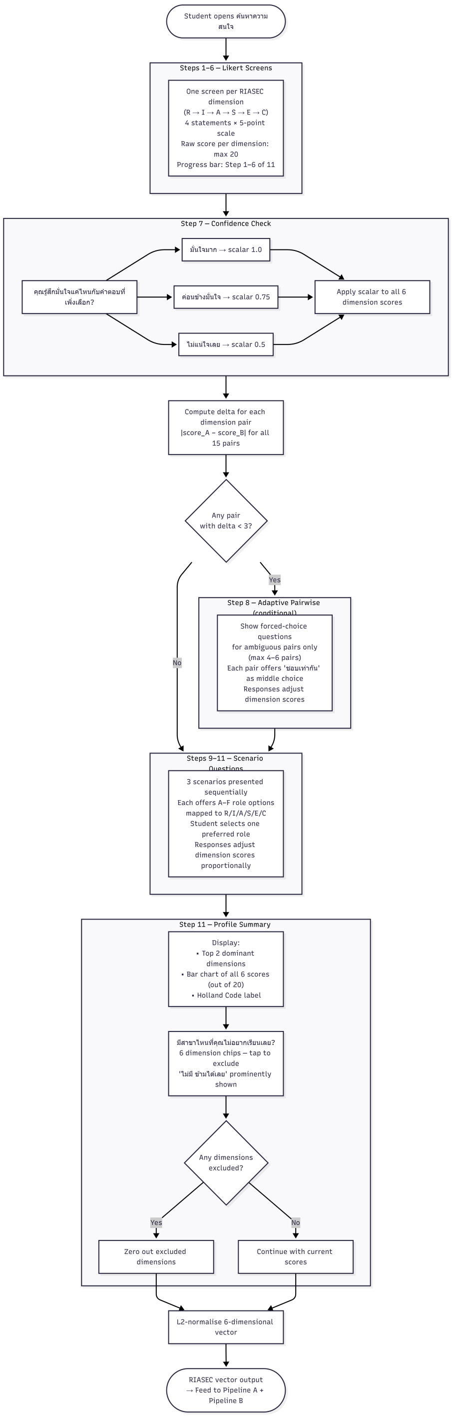

Figure 4.6: Interest Elicitation Flow (UC-01) — 12-Step Adaptive Process

Chapter 5

Software Development

5.1 Schedule

Development is organized into four phases across April 2026 and January–March 2027. P1 Foundation runs April 1–12, P2 Core AI POC runs April 6–30, P3 MVP Feature Completion runs January 2027, and P4 Polish & Evaluation runs February–March 2027. By the end of April 2026, the current proof of concept demonstrates the data ingestion pipeline, grounded RAG chatbot, and Program Explorer over indexed curriculum content. Recommendation, explanation, and PLO/profile visualisation remain MVP features to be completed and evaluated during P3–P4.

Table 5.1: Project Schedule

Period	Phase	Deliverables
Apr 1–5, 2026	P1 · Foundation — Set up Repo	Repository setup, project structure, development environment
Apr 1–5, 2026	P1 · Foundation — Data Collection	Receive มคอ.2 and TCAS admission data from KU faculty, download O*NET dataset
Apr 6–12, 2026	P1 · Foundation — Data Ingestion	PDF extraction, chunking, embedding into Supabase pgvector; graph schema exploration for Phase 2
Apr 6–30, 2026	P2 · Core AI POC — RAG Chatbot	RAG pipeline over indexed curriculum and TCAS/fee data, grounded answer generation, source citations
Apr 6–30, 2026	P2 · Core AI POC — Program Explorer	Semantic search and browsing over indexed program content and metadata using Supabase/pgvector
Apr 6–30, 2026	P2 · Core AI POC — Demo Integration	Working proof-of-concept demo combining ingestion output, Program Explorer, and RAG chatbot
Jan 1–31, 2027	P3 · MVP Features — Interest Discovery	RIASEC elicitation flow, profile scoring, confidence adjustment, and profile summary
Jan 1–31, 2027	P3 · MVP Features — Program Detail Pages	Program detail pages, static PLO/profile chart, and available TCAS information
Jan 1–31, 2027	P3 · Phase 2 Prototype — Portfolio Coach	Optional prototype UI only; excluded from MVP pass/fail evaluation
Jan 31, 2027	P3 · MVP Features — Complete Recommendation	Complete simplified Pipeline B2 recommendation, evidence-grounded explanation, and Program Explorer refinement
Feb 1–28, 2027	P4 · Polish & Eval — UI/UX Polish	UI/UX polish, Thai/English language switching, mobile-first refinement
Feb 1–28, 2027	P4 · Polish & Eval — Cross-feature Integration	Chatbot + Explorer + Recommendation + Program Detail integration
Feb 1–28, 2027	P4 · Polish & Eval — User Testing	RAGAS plus custom TCAS/fee regression, MRR/NDCG test set, SUS user testing with 10–15 students
Mar 1–31, 2027	P4 · Polish & Eval — Performance Testing	Performance benchmarking against all success metric targets
Mar 1–31, 2027	P4 · Polish & Eval — Bug Fixing & Optimization	Bug fixes, optimization, stability improvements
Mar 1–31, 2027	P4 · Polish & Eval — Final Documentation	Final report, demo preparation, project documentation

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Appendix A

Appendix A: RIASEC Elicitation Question Bank and Scoring Logic

This appendix documents the full question bank used by the 12-step interest discovery interface (UC-01) and the scoring rules applied at each stage. After all steps are complete, the 6-dimensional RIASEC vector is L2-normalised before use in the recommendation pipeline.

Adaptive threshold note: Pairwise questions (Table A.2) are shown only for dimension pairs where $|\text{score}_A - \text{score}_B| < 3$ after confidence scaling. A maximum of 4–6 pairs are presented per session.

Table A.1: Likert Question Bank (4 statements per RIASEC dimension, 5-point scale, max score 20)

Dimension	Q	Statement (Thai)
Realistic (R)	1	ฉันชอบงานที่ได้ใช้มือและร่างกาย เช่น ประกอบ ซ่อม หรือสร้างสิ่งของ
	2	ฉันสนุกกับการทำงานกับเครื่องมือ เครื่องจักร หรืออุปกรณ์ต่าง ๆ
	3	ฉันชอบแก้ปัญหาที่จับต้องได้มากกว่าการคิดเชิงนามธรรม
	4	ฉันรู้สึกสบายใจเมื่อได้ทำงานกลางแจ้งหรือในพื้นที่จริง
Investigative (I)	1	ฉันชอบวิเคราะห์ข้อมูลและหาคำตอบจากหลักฐาน
	2	ฉันสนุกกับการตั้งสมมติฐานและทดสอบแนวคิดใหม่ ๆ
	3	ฉันชอบอ่านหรือค้นคว้าเรื่องที่ฉันสนใจอย่างลึกซึ้ง
	4	ฉันรู้สึกสนุกเมื่อได้แก้ปัญหาที่ซับซ้อนหรือต้องใช้ตรรกะ
Artistic (A)	1	ฉันชอบแสดงความคิดสร้างสรรค์ผ่านงานศิลปะ ดนตรี หรือการเขียน
	2	ฉันมักมองเห็นโอกาสในการออกแบบหรือปรับแต่งสิ่งต่าง ๆ ให้สวยงามขึ้น
	3	ฉันรู้สึกสบายใจในสภาพแวดล้อมที่เปิดโอกาสให้แสดงตัวตน
	4	ฉันชอบงานที่ไม่มีกฎตายตัวและมีอิสระในการสร้างสรรค์
Social (S)	1	ฉันรู้สึกมีพลังเมื่อได้ช่วยเหลือหรืออธิบายให้คนอื่นเข้าใจ
	2	ฉันชอบทำงานเป็นทีมและสร้างความสัมพันธ์กับผู้อื่น
	3	ฉันสนใจปัญหาสังคมและอยากมีส่วนร่วมในการแก้ไข
	4	ฉันชอบให้คำปรึกษาหรือสนับสนุนเพื่อน ๆ เมื่อพวกเขาต้องการ
Enterprising (E)	1	ฉันชอบโน้มน้าวคนอื่นและนำเสนอแนวคิดของตัวเอง
	2	ฉันสนุกกับการวางแผนโครงการและนำทีมให้บรรลุเป้าหมาย
	3	ฉันรู้สึกตื่นเต้นกับความท้าทายและโอกาสทางธุรกิจ
	4	ฉันชอบตัดสินใจและรับผิดชอบผลลัพธ์ที่เกิดขึ้น
Conventional (C)	1	ฉันชอบงานที่มีขั้นตอนชัดเจนและทำตามระบบได้
	2	ฉันรู้สึกสบายใจเมื่องานมีความแม่นยำและตรวจสอบได้
	3	ฉันชอบจัดระเบียบข้อมูล แฟ้ม หรือตารางให้เป็นระบบ
	4	ฉันทำงานได้ดีในสภาพแวดล้อมที่มีกฎและความคาดหวังที่ชัดเจน

Table A.2: Pairwise Question Bank (6 adjacent dimension pairs)

Pair	Option A	Option B
R vs. I	ออกแบบและสร้างเครื่องมือทางกายภาพ	วิเคราะห์ข้อมูลและพัฒนาทฤษฎีใหม่
I vs. A	ค้นคว้าและทดสอบสมมติฐานทางวิทยาศาสตร์	สร้างสรรค์ผลงานศิลปะหรือสื่อสารสร้างสรรค์
A vs. S	ออกแบบประสบการณ์หรือผลงานที่สวยงาม	สอน ให้คำปรึกษา หรือดูแลผู้อื่น
S vs. E	ทำงานสนับสนุนและช่วยเหลือชุมชน	นำทีมและขับเคลื่อนองค์กรให้เติบโต
E vs. C	บริหารโครงการและสร้างธุรกิจใหม่	จัดการระบบและตรวจสอบความถูกต้อง
R vs. C	งานภาคสนามที่ต้องใช้ทักษะเชิงปฏิบัติ	งานสำนักงานที่เน้นความเป็นระบบและความแม่นยำ

Table A.3: Scenario Question Bank (3 scenarios × 6 role options with RIASEC mapping)

Scenario	Setting	Role option	RIASEC
1	งานแฮกกาธอน 48 ชั่วโมง	A — ออกแบบและสร้าง prototype ที่ใช้งานได้จริง	R
		B — วิเคราะห์ข้อมูลและหาโซลูชันที่ดีที่สุด	I
		C — ออกแบบ UI/UX ให้น่าใช้งานและสวยงาม	A
		D — ประสานงานทีมและดูแลสมาชิกให้ทำงานได้ดี	S
		E — นำเสนอผลงานต่อกรรมการและโน้มน้าวให้ชนะ	E
		F — จัดการตารางงาน งบประมาณ และเอกสารทีม	C
2	โปรเจกต์ชุมชน	A — ก่อสร้างหรือซ่อมแซมโครงสร้างพื้นฐาน	R
		B — ศึกษาปัญหาและออกแบบแนวทางแก้ไขเชิงระบบ	I
		C — ผลิตสื่อสร้างสรรค์เพื่อสร้างการรับรู้	A
		D — พูดคุยกับชาวบ้านและรวบรวมความต้องการ	S
		E — ระดมทุนและหาพาร์ทเนอร์สนับสนุนโครงการ	E
		F — ติดตามความคืบหน้าและรายงานผลให้ถูกต้อง	C
3	สตาร์ทอัพด้านสุขภาพ	A — พัฒนาอุปกรณ์หรือระบบฮาร์ดแวร์ที่ใช้งานได้	R
		B — วิจัยและทดสอบประสิทธิภาพของผลิตภัณฑ์	I
		C — สร้างแบรนด์และออกแบบประสบการณ์ผู้ใช้	A
		D — ฝึกอบรมและสนับสนุนผู้ใช้งานในชุมชน	S
		E — วางกลยุทธ์ธุรกิจและขยายตลาด	E
		F — ดูแลการปฏิบัติตามกฎระเบียบและมาตรฐาน	C

Appendix B

Appendix B: About LaTeX

LaTeX (stylized as LaTeX) is a software system for typesetting documents. LaTeX markup describes the content and layout of the document, as opposed to the formatted text found in WYSIWYG word processors like Google Docs, LibreOffice Writer, and Microsoft Word. The writer uses markup tagging conventions to define the general structure of a document, to stylize text throughout a document (such as bold and italics), and to add citations and cross-references.

LaTeX is widely used in academia for the communication and publication of scientific documents and technical note-taking in many fields, owing partially to its support for complex mathematical notation. It also has a prominent role in the preparation and publication of books and articles that contain complex multilingual materials, such as Arabic and Greek.

Overleaf has also provided a 30-minute guide on how you can get started on using LaTeX. [14]